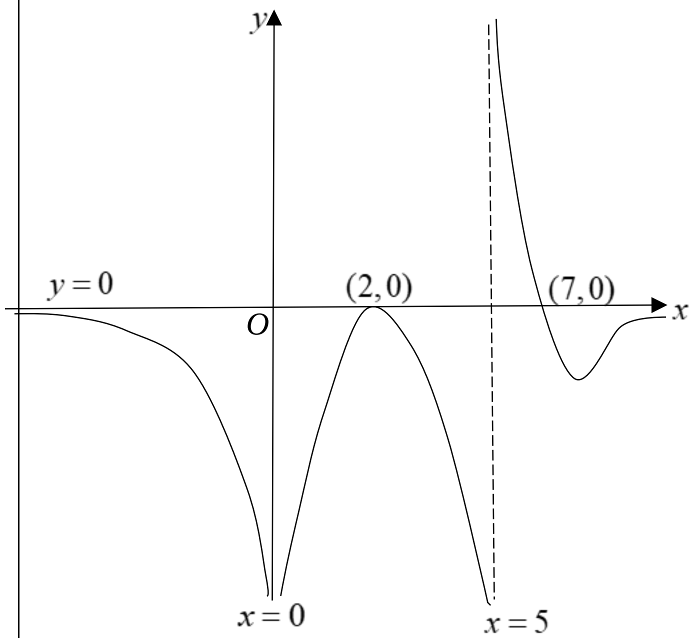
Promo Practice Paper 4 [HCI 2022] Solutions

No.	Suggested Solutions	
1		
2	Let A, C and S be the number of adults, children and senior citizens in the local family. $27A+18.4C+15S=301.2$ ------------- (1) $39A+29.5C+39S=478.50$ ----------- (2) $A-4S=0$ ------------------------------- (3) By GC, $A=8$ $C=3$ $S=2$ $\therefore A=8,\ C=3,\ S=2$ The family consists of 8 adults, 3 children and 2 senior citizens.	
3(i)	$kxe^{y}+ke^{x}=y^{2}+k^{2}$ Differentiating with respect to x,	

No.	Suggested Solutions	
	$ke^y + kxe^y \frac{dy}{dx} + ke^x = 2y\frac{dy}{dx}$ $\frac{dy}{dx}\left(2y - kxe^y\right) = ke^x + ke^y$ $\frac{dy}{dx} = \frac{ke^x + ke^y}{2y - kxe^y}$	
3 (ii)	For tangent to the curve to be parallel to the x-axis, $\frac{dy}{dx} = 0$ $\therefore ke^x + ke^y = 0$ For all x, y, $e^x > 0$, $e^y > 0$ Hence, $ke^x + ke^y > 0$ $\therefore \frac{dy}{dx} \neq 0$ for all $x, y \in \mathbb{R}$ Therefore, there is no point where the tangent is parallel to the x-axis.	
4(i)	$\left(\underset{\sim}{r} - \underset{\sim}{p}\right) \times \left(\underset{\sim}{p} - \underset{\sim}{q}\right) = \underset{\sim}{0}$ $\Rightarrow \left(\overrightarrow{PR}\right) \times \left(\overrightarrow{QP}\right) = \underset{\sim}{0}$ $\Rightarrow \overrightarrow{PR} // \overrightarrow{QP}$ Common point P, $\therefore P, Q$ and R are collinear **Alternative method** $\left(\underset{\sim}{r} - \underset{\sim}{p}\right) = \lambda\left(\underset{\sim}{p} - \underset{\sim}{q}\right), \quad \lambda \in \mathbb{R}$ $\underset{\sim}{r} = \underset{\sim}{p} + \lambda\left(\underset{\sim}{p} - \underset{\sim}{q}\right), \quad \lambda \in \mathbb{R}$ $\therefore R$ is a point on line passing through P and Q	
4(ii)	Since $\left\lvert\underset{\sim}{q}\right\rvert = 1$ $\left\lvert\underset{\sim}{q} \bullet \left(\underset{\sim}{r} - \underset{\sim}{s}\right)\right\rvert$ $= \left\lvert\overrightarrow{SR} \bullet \underset{\sim}{q}\right\rvert$ $\therefore \left\lvert\underset{\sim}{q} \bullet \left(\underset{\sim}{r} - \underset{\sim}{s}\right)\right\rvert$ is the length of projection of $\overrightarrow{SR}$ onto $\underset{\sim}{q}$	

No.	Suggested Solutions	
4(iii)	$\underset{\sim}{p}=\begin{pmatrix}-1\\2\\4\end{pmatrix}$ $\underset{\sim}{q}=-\frac{1}{\sqrt{(-1)^2+2^2+4^2}}\begin{pmatrix}-1\\2\\4\end{pmatrix}=-\frac{1}{\sqrt{21}}\begin{pmatrix}-1\\2\\4\end{pmatrix}$	
5	**MI Promo 9758/2020/PU2/P1/Q8a**	
(a) (i)	$ab^*=(3-\mathrm{i})(5-2\mathrm{i})$ $=15-6\mathrm{i}-5\mathrm{i}+2\mathrm{i}^2$ $=15-6\mathrm{i}-5\mathrm{i}-2$ $=13-11\mathrm{i}$	
(a)(ii)	$\frac{b}{a^*}=\frac{5+2\mathrm{i}}{3+\mathrm{i}}\times\frac{3-\mathrm{i}}{3-\mathrm{i}}$ $=\frac{15-5\mathrm{i}+6\mathrm{i}-2\mathrm{i}^2}{9-\mathrm{i}^2}$ $=\frac{15-5\mathrm{i}+6\mathrm{i}+2}{9+1}$ $=\frac{1}{10}(17+\mathrm{i})$	
6	**RVHS Promo 9758/2020/Q3**	
(i)	**Method 1:** $\tan^{-1}(y)=1-\mathrm{e}^{-x}$ Differentiate implicitly w.r.t. x, $\frac{1}{1+y^2}\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)=\mathrm{e}^{-x}$ $\mathrm{e}^{x}\frac{\mathrm{d}y}{\mathrm{d}x}=1+y^2$. (Shown) **Or** **Method 2:** $\tan^{-1}(y)=1-\mathrm{e}^{-x}$ $y=\tan\left(1-\mathrm{e}^{-x}\right)$ Differentiate w.r.t. x,	

No.	Suggested Solutions	
	$\frac{\mathrm{d}y}{\mathrm{d}x}=\mathrm{e}^{-x}\sec^2\left(1-\mathrm{e}^{-x}\right)$ $\mathrm{e}^x\frac{\mathrm{d}y}{\mathrm{d}x}=1+\tan^2\left(1-\mathrm{e}^{-x}\right)$ $\mathrm{e}^x\frac{\mathrm{d}y}{\mathrm{d}x}=1+y^2$. (Shown)	
(ii)	$\mathrm{e}^x\frac{\mathrm{d}y}{\mathrm{d}x}=1+y^2$ (1) Differentiate implicitly w.r.t. x, $\mathrm{e}^x\frac{\mathrm{d}y}{\mathrm{d}x}+\mathrm{e}^x\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=2y\frac{\mathrm{d}y}{\mathrm{d}x}$ (2) When $x=0, \tan^{-1}(y)=1-\mathrm{e}^{-0}=0 \Rightarrow y=\tan(0)=0$, From (1): $\mathrm{e}^0\frac{\mathrm{d}y}{\mathrm{d}x}=1+0^2 \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x}=1$, From (2): $1+0^2+\mathrm{e}^0\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=2(0)(1) \Rightarrow \frac{\mathrm{d}^2y}{\mathrm{d}x^2}=-1$, Hence, $y=0+(1)x+\left(\frac{-1}{2!}\right)x^2+\dots$ $=x-\frac{1}{2}x^2+\dots$	
(iii)	$\mathrm{e}^x\frac{\mathrm{d}y}{\mathrm{d}x}=1+y^2$ $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1+y^2}{\mathrm{e}^x}$. Hence, $\frac{1+y^2}{\mathrm{e}^x}=\frac{\mathrm{d}y}{\mathrm{d}x}$ $=\frac{\mathrm{d}}{\mathrm{d}x}\left(x-\frac{1}{2}x^2+\dots\right)$ $=1-x+\dots$	

No.	Suggested Solutions	

7 (i) $x^2 + y^2 = 16$

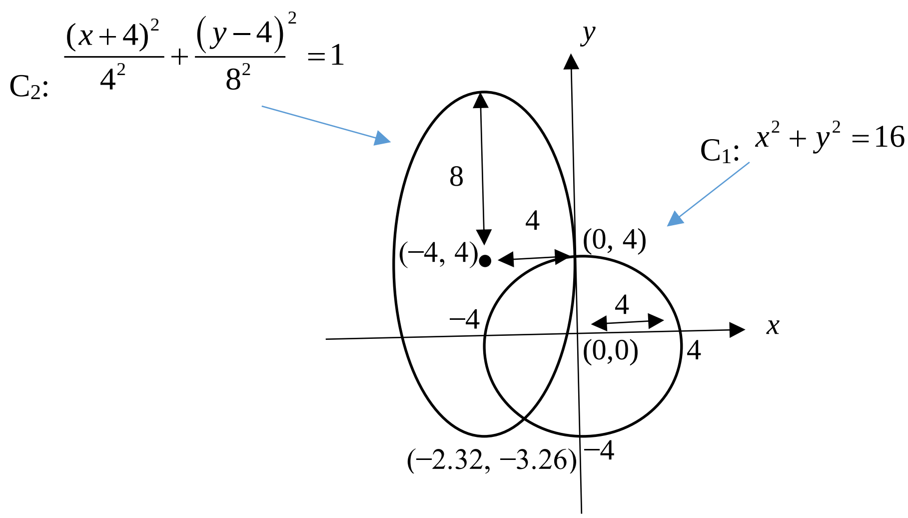
Transformation A, replace x with x+4: $(x+4)^2 + y^2 = 16$

Transformation B, replace y with y–2: $(x+4)^2 + (y-2)^2 = 16$

Transformation C, replace y with $\frac{y}{2}$: $(x+4)^2 + \left(\frac{y}{2}-2\right)^2 = 16$

$$\frac{(x+4)^2}{4^2} + \frac{(y-4)^2}{8^2} = 1$$

(ii)



(iii) Intersection points are (–2.32, –3.26) and (0, 4)

(iv) $$\int_{-3.259216}^{4} \left(\left(-4 + 4\sqrt{\left(1 - \frac{(y-4)^2}{8^2}\right)} \right) - \left(-\sqrt{16 - y^2} \right) \right) dy = 19.2$$

8(i) $S_n = e^2 - (-2)^n \left(e^{2-n}\right)$

$T_1 = S_1$

$= e^2 - (-2)\left(e^{2-1}\right)$

$= e^2 + 2e$

No.	Suggested Solutions	
8(ii)	$T_n = S_n - S_{n-1}$ $= \left[e^2 - (-2)^n (e^{2-n})\right] - \left[e^2 - (-2)^{n-1}(e^{2-n+1})\right]$ $= (-2)^{n-1}(e^{2-n+1}) - (-2)^n (e^{2-n})$ $= (-2)^{n-1}(e^{2-n})\left[e - (-2)\right]$ $= (-2)^{n-1}(e^{2-n})\left[e+2\right]$ Since $\frac{T_{n+1}}{T_n} = \frac{(-2)^n (e^{1-n})[e+2]}{(-2)^{n-1}(e^{2-n})[e+2]} = -\frac{2}{e}$ is a constant This is a Geometric Series with common ratio $-\frac{2}{e}$.	
8(iii)	Since $\lvert r \rvert = \left\lvert -\frac{2}{e} \right\rvert = 0.736 \text{ (3 s.f.)} < 1$ Sum to infinity exits. $S = \frac{a}{1-r}$ $S = \frac{e(2+e)}{1-\left(-\frac{2}{e}\right)}$ $= \frac{e(2+e)}{\left(\frac{e+2}{e}\right)}$ $= e^2$	
9(i)	When C crosses the x-axis , $y = 0$ $y = 2\sin\theta + 1 = 0$ $\sin\theta = -\frac{1}{2}$ $\theta = -\frac{\pi}{6}, -\frac{5\pi}{6}$ since $-\pi \le \theta \le \pi$	

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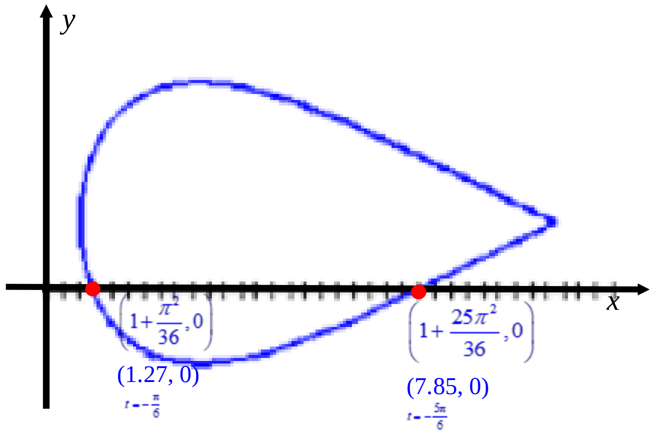
(ii)</td>
<td>
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$\frac{\mathrm{d}x}{\mathrm{d}\theta} = 2\theta$, $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 2\cos\theta$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}\theta} \div \frac{\mathrm{d}\theta}{\mathrm{d}x} = \frac{\cos\theta}{\theta}$

$\Rightarrow$ At P, $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\cos p}{p}$

Equation of tangent at P :

$y - (2\sin p + 1) = \frac{\cos p}{p}\left(x - p^2 - 1\right)$

$y = \frac{x\cos p}{p} - \frac{\left(p^2+1\right)\cos p}{p} + 2\sin p + 1$

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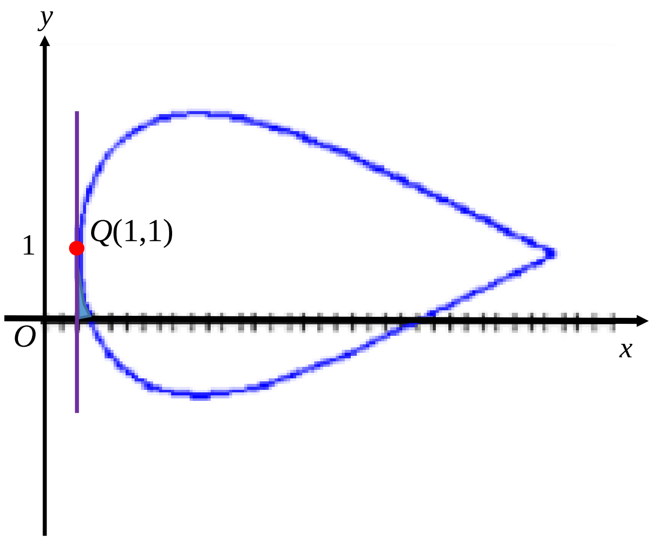
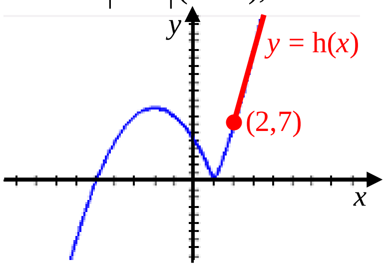
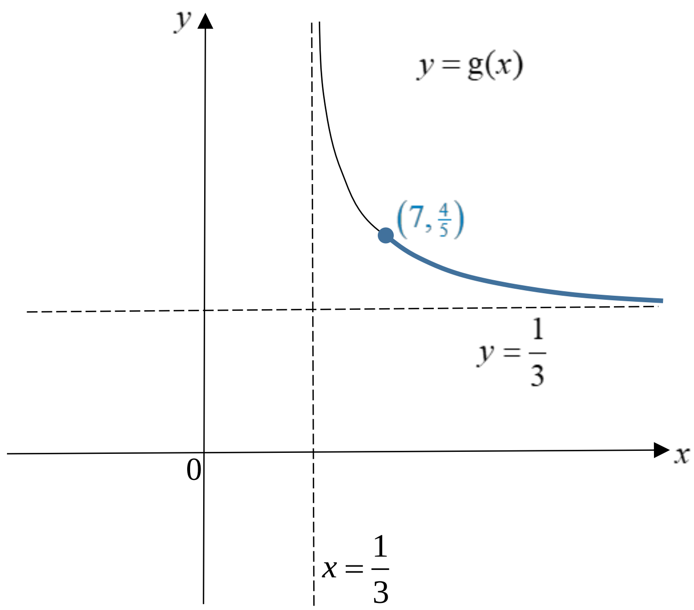
(iv)	 Tangent parallel to the y-axis: $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\cos\theta}{\theta}$ is undefined at $\theta=0$, $Q\ (1,1)$	
	Method 1: Using parametric equations Required Area $=\int_{1}^{1+\frac{\pi^2}{36}} y\,\mathrm{d}x$ $=\int_{0}^{-\frac{\pi}{6}}(1+2\sin\theta)(2\theta)\,\mathrm{d}\theta$ = 0.087955042 = 0.0880 units^2 (3 s.f.)	
	Method 2: Using Cartesian equation $x=\theta^2+1$ $\theta=\pm\sqrt{x-1}$, $y=2\sin\theta+1$ $y=2\sin\left(-\sqrt{x-1}\right)+1$ Required Area	

	$= \int_1^{1+\frac{\pi^2}{36}} y \, \mathrm{d}x$ $= \int_1^{1+\frac{\pi^2}{36}} 2\sin\left(-\sqrt{x-1}\right)+1 \, \mathrm{d}x$ = 0.087954559 = 0.0880 units^2 (3 s.f.)	
10(i)	**<u>Method 1</u>** $\mathrm{g}(1)=5$ $\frac{1+a}{3+b}=5 \Rightarrow 1+a=15+5b \Rightarrow a-5b=14$ Since g is self-inverse, $\mathrm{g}(5)=1$. $\frac{5+a}{15+b}=1 \Rightarrow 5+a=15+b \Rightarrow a-b=10$ Using GC, $a=9$, $b=-1$.	
	<u>Method 2</u> Let $y=\frac{x+a}{3x+b}$ $3xy+by=x+a$ $\Rightarrow x=\frac{a-by}{3y-1}$ $\mathrm{g}^{-1}(x)=\frac{a-bx}{3x-1}$ Since g is self-inverse, $\frac{a-bx}{3x-1}=\frac{x+a}{3x+b}$ $b=-1$ $\mathrm{g}(1)=5$ $\frac{1+a}{3+b}=5 \Rightarrow 1+a=15+5b \Rightarrow a-5b=14$ $a=14-5=9$	
	<u>Method 3</u> $\mathrm{g}(x)=\frac{x+a}{3x+b}=\frac{\frac{1}{3}(3x+b)-\frac{1}{3}b+a}{3x+b}=\frac{1}{3}-\frac{b-3a}{3(3x+b)}$ Since $y=\frac{1}{3}$ is the horizontal asymptote, by symmetry $x=\frac{1}{3}$ is the vertical asymptote. $\therefore b=-1$ $\mathrm{g}(1)=5$	

	$\frac{1+a}{3+b}=5 \Rightarrow 1+a=15+5b \Rightarrow a-5b=14$ $a=14-5=9$	
(ii)	Since $\mathrm{g}(x)=\mathrm{g}^{-1}(x)$, $\mathrm{g}^2(x)=x$ $\mathrm{g}^3(x)=g\left[\mathrm{g}^2(x)\right]=\mathrm{g}(x)$ $\therefore \mathrm{g}^{2021}(1)=g\left[\mathrm{g}^{2020}(1)\right]=\mathrm{g}(1)=5$	
(iii)	$\mathrm{h}: x \mapsto \lvert 1-x\rvert(x+5)$, for $x \in \mathbb{R}$, $x \geq 2$.  Since $x \geq 2$, $\lvert 1-x\rvert=x-1$ $\therefore \mathrm{h}(x)=(x-1)(x+5)$ Let $y=x^2+4x-5$ $x^2+4x-5-y=0$ $x=\frac{-4 \pm \sqrt{16-4(-5-y)}}{2}$ $=\frac{-4 \pm \sqrt{36+4y}}{2}$ $=-2 \pm \sqrt{9+y}$ Since $x \geq 2$, $x=-2+\sqrt{9+y}$ $\mathrm{h}^{-1}: x \mapsto \sqrt{9+x}-2, \quad x \in \mathbb{R}, x \geq 7.$	

(iv	 $\mathrm{R_h}=[7,\infty)$ and $\mathrm{D_g}=\left(\frac{1}{3},\infty\right)$ Since $\mathrm{R_h}\subseteq\mathrm{D_g}$ $\therefore$ gh exits $\mathrm{R_{gh}}=\left(\frac{1}{3},\frac{4}{5}\right]$	
11(i)	<u>Method 1</u>: (solve algebraically) When $y=-1$, $-1=\frac{(1-x)(x-2)}{x^2-2x-8}$ $-x^2+2x+8=-x^2+3x-2$ $\therefore\quad x=10$ <u>Method 2</u>: (solve graphically using GC) Using GC, $\therefore\ x=10$	

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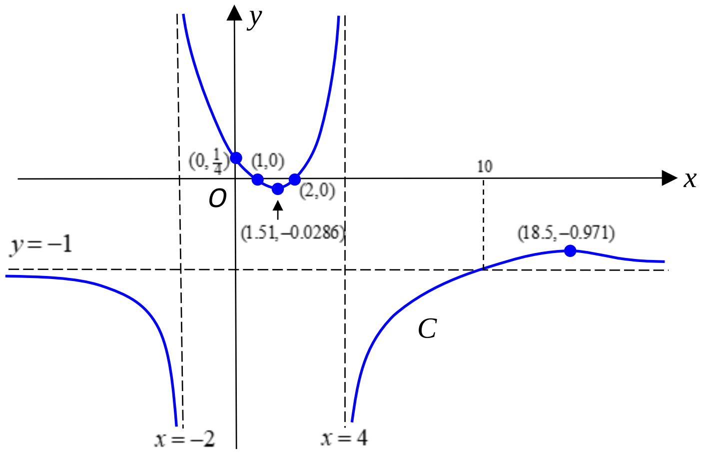
$$y=\frac{(1-x)(x-2)}{x^2-2x-8}$$
$$=\frac{-x^2+3x-2}{x^2-2x-8}$$
$$=-1+\frac{x-10}{(x+2)(x-4)}$$

$$\begin{array}{r} -1 \\ x^2-2x-8\overline{)-x^2+3x-2} \\ \underline{-x^2+2x+8} \\ x-10 \end{array}$$



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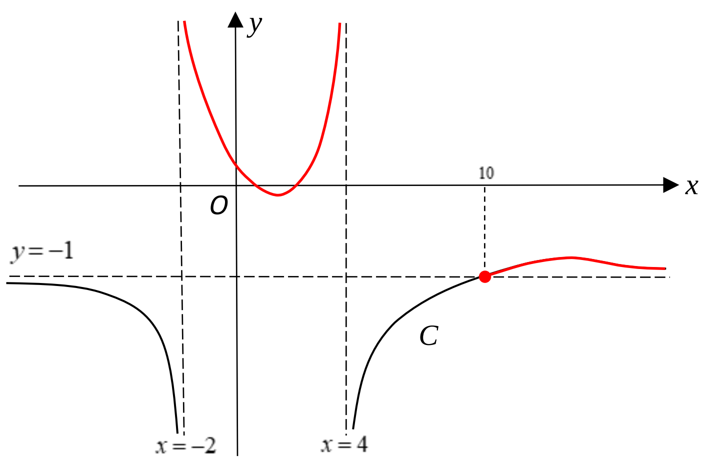
<u>Method 1</u>: [using graph in part (ii)]

$\frac{-x^2+3x-2}{x^2-2x-8}\geq -1 \Rightarrow$ Curve C ≥ -1



$\therefore$ from graph in (ii), $-2<x<4$ or $x\geq 10$

<u>Method 2</u>: (solve inequality directly)

$\frac{-x^2+3x-2}{x^2-2x-8}\geq -1$

$-1+\frac{x-10}{(x+2)(x-4)}\geq -1$

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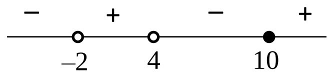
$\frac{x-10}{(x+2)(x-4)} \geq 0$

Use simplified form after long division from part (ii)

$\therefore\ -2 < x < 4$ or $x \geq 10$



(iv)

Replace x with $\frac{1}{x}$ in (iii): $\frac{-(\frac{1}{x})^2+3(\frac{1}{x})-2}{(\frac{1}{x})^2-2(\frac{1}{x})-8} \geq -1$

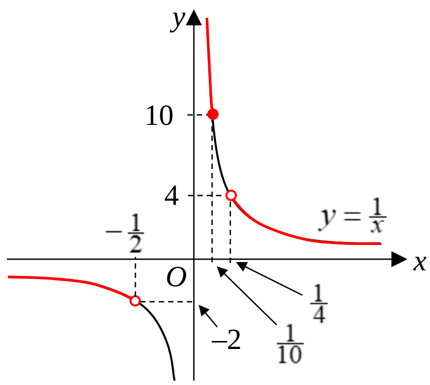
$$\frac{\frac{-1+3x-2x^2}{x^2}}{\frac{1-2x-8x^2}{x^2}} \geq -1$$

$$\frac{-1+3x-2x^2}{1-2x-8x^2} \geq -1$$

Hence using result in (iii):

$-2 < \frac{1}{x} < 4$ or $\frac{1}{x} \geq 10$

<u>Method 1</u>: (using graph of $y = \frac{1}{x}$)



From graph,

$x < -\frac{1}{2}$ or $x > \frac{1}{4}$ $\quad 0 < x \leq \frac{1}{10}$

Note that $x = 0$ is also a solution to $\frac{-1+3x-2x^2}{1-2x-8x^2} \geq -1$

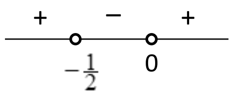
Hence the final solution is $x < -\frac{1}{2}$ or $x > \frac{1}{4}$ or $0 \leq x \leq \frac{1}{10}$

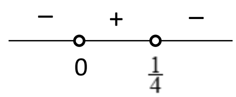
<u>Method 2</u>: (solving algebraically)

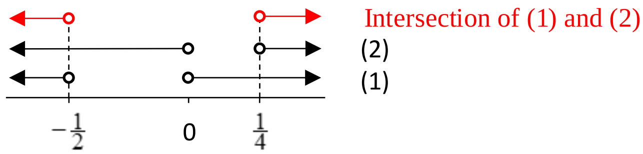
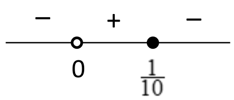
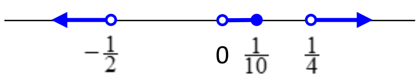
For $-2 < \frac{1}{x} < 4$:

$-2 < \frac{1}{x}$	and	$\frac{1}{x} < 4$
$\frac{1}{x} + 2 > 0$		$\frac{1}{x} - 4 < 0$
$\frac{1+2x}{x} > 0$		$\frac{1-4x}{x} < 0$

$\therefore\ x < -\frac{1}{2}$ or $x > 0$ …(1) and $\therefore\ x < 0$ or $x > \frac{1}{4}$ …(2)





	Combining (1) and (2), ie. taking intersection:  Hence $x<-\frac{1}{2}$ or $x>\frac{1}{4}$ (*) For $\frac{1}{x}\geq 10$: $\frac{1}{x}-10\geq 0$ $\frac{1-10x}{x}\geq 0$  $\therefore\ 0<x\leq\frac{1}{10}$ (**) Hence combining (*) or (**), ie. taking union:  $\therefore\ x<-\frac{1}{2}$ or $0<x\leq\frac{1}{10}$ or $x>\frac{1}{4}$ Note that $x=0$ is also a solution to $\frac{-1+3x-2x^2}{1-2x-8x^2}\geq -1$ Hence the final solution is $x<-\frac{1}{2}$ or $x>\frac{1}{4}$ or $0\leq x\leq\frac{1}{10}$	
12 **(i)**	$V=\pi r^2h+\frac{1}{2}\left[\frac{4}{3}\pi\left(\frac{r}{2}\right)^3\right]$ $V=\pi r^2h+\frac{2}{3}\pi\left(\frac{r^3}{8}\right)$ $h=\frac{V}{\pi r^2}-\frac{r}{12}$	
(ii)	Let C be the total cost of manufacturing the prototype. $C=8[\text{Cylinder's: Base area + Curved Surface + Top}]$ $+3[\text{Curved Surface of hemisphere}]$	•

	$C=8\left[\pi r^2+2\pi rh+\left\{\pi r^2-\pi\left(\frac{r}{2}\right)^2\right\}\right]+3\times\frac{1}{2}\left[4\pi\left(\frac{r}{2}\right)^2\right]$ $C=8\left[\pi\left(2r^2-\frac{r^2}{4}\right)+2\pi rh\right]+3\left[2\pi\left(\frac{r^2}{4}\right)\right]$ $=8\left[\frac{7\pi r^2}{4}+2\pi r\left(\frac{V}{\pi r^2}-\frac{r}{12}\right)\right]+\frac{3\pi r^2}{2}$ $=8\left[\frac{19\pi r^2}{12}+\frac{2V}{r}\right]+\frac{3\pi r^2}{2}$ $=\frac{16V}{r}+\frac{85\pi r^2}{6}$ (shown)	
	$\frac{\mathrm{d}C}{\mathrm{d}r}=\frac{85}{3}\pi r-\frac{16V}{r^2}$ $\frac{\mathrm{d}C}{\mathrm{d}r}=\frac{1}{r^2}\left(\frac{85}{3}\pi r^3-16V\right)$ For stationary value, $\frac{\mathrm{d}C}{\mathrm{d}r}=0$ $r^3=\frac{48V}{85\pi}$ $r=\sqrt[3]{\frac{48V}{85\pi}}$	
	Using 2nd Derivative Test: $\frac{\mathrm{d}^2C}{\mathrm{d}r^2}=\frac{85}{3}\pi+\frac{32V}{r^3}=\frac{85}{3}\pi+\frac{32V}{\frac{48V}{85\pi}}=85\pi>0$ Thus, C is a minimum when $r=\sqrt[3]{\frac{48V}{85\pi}}$.	
	$h=\frac{V}{\pi r^2}-\frac{r}{12}$	

$$\frac{h}{r} = \frac{V}{\pi r^3} - \frac{1}{12}$$

$$\frac{h}{r} = \frac{V}{\pi\left(\frac{48V}{85\pi}\right)} - \frac{1}{12}$$

$$\frac{h}{r} = \frac{85}{48} - \frac{1}{12} = \frac{27}{16}$$

$$\frac{\mathrm{d}C}{\mathrm{d}r} = \frac{85}{3}\pi r - \frac{16V}{r^2}$$

$$\frac{\mathrm{d}C}{\mathrm{d}r} = \frac{1}{r^2}\left(\frac{85}{3}\pi r^3 - 16V\right)$$

$$\frac{\mathrm{d}C}{\mathrm{d}r} = 0$$

Using 1st Derivative Test:

r	$\sqrt[3]{\frac{48V}{85\pi}}^{-}$	$\sqrt[3]{\frac{48V}{85\pi}}$	$\sqrt[3]{\frac{48V}{85\pi}}^{+}$
$\frac{\mathrm{d}C}{\mathrm{d}r}$	–	0	+
Shape	\	–	/

(iii) After 5 seconds, amount of perfume leaked is

$1.5 \times 5 = 7.5 \text{ cm}^3$

Total volume of hemisphere is

$$\frac{2}{3}\pi\left(\frac{3}{2}\right)^3 = \frac{9\pi}{4} = 7.06858 \text{ cm}^3 < 7.5$$

After 5 seconds, perfume is in the bottom segment, i.e. cylinder.

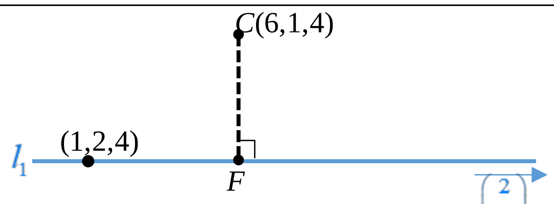
$V = \pi r^2 h = 9\pi h$

$$\frac{\mathrm{d}V}{\mathrm{d}h} = 9\pi$$

$$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{\mathrm{d}h}{\mathrm{d}V} \times \frac{\mathrm{d}V}{\mathrm{d}t} = \frac{1}{9\pi} \times (-1.5) = -\frac{1}{6\pi} \text{ cm / s}$$

Height of perfume decreases at $\frac{1}{6\pi}$ cm / s.

13(i)



Let the foot of perpendicular be F.

Since F lies on l_1,

$$\overrightarrow{OF}=\begin{pmatrix}1\\2\\4\end{pmatrix}+\lambda\begin{pmatrix}2\\-3\\0\end{pmatrix} \text{ for some } \lambda$$

$$\overrightarrow{CF}=\begin{pmatrix}1\\2\\4\end{pmatrix}+\lambda\begin{pmatrix}2\\-3\\0\end{pmatrix}-\begin{pmatrix}6\\1\\4\end{pmatrix}=\begin{pmatrix}-5+2\lambda\\1-3\lambda\\0\end{pmatrix}$$

Since $\overrightarrow{CF}\perp l_1$

$$\begin{pmatrix}-5+2\lambda\\1-3\lambda\\0\end{pmatrix}\bullet\begin{pmatrix}2\\-3\\0\end{pmatrix}=0$$

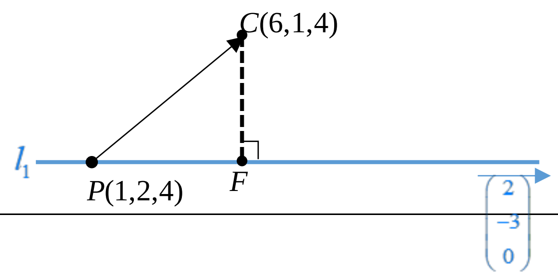
$$-10+4\lambda-3+9\lambda=0$$

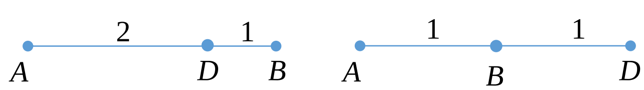
$$\lambda=1$$

$$\overrightarrow{OF}=\begin{pmatrix}1\\2\\4\end{pmatrix}+1\begin{pmatrix}2\\-3\\0\end{pmatrix}=\begin{pmatrix}3\\-1\\4\end{pmatrix}$$

$\therefore F(3,-1,4)$

Alternative Solution (Projection Vector Method)



	$\overrightarrow{PC}=\begin{pmatrix}6\\1\\4\end{pmatrix}-\begin{pmatrix}1\\2\\4\end{pmatrix}=\begin{pmatrix}5\\-1\\0\end{pmatrix}$ $\overrightarrow{PF}=\dfrac{\overrightarrow{PC}\cdot\begin{pmatrix}2\\-3\\0\end{pmatrix}}{\left(\sqrt{4+9}\right)^2}\begin{pmatrix}2\\-3\\0\end{pmatrix}$ $=\begin{pmatrix}2\\-3\\0\end{pmatrix}$ $\overrightarrow{OF}=\overrightarrow{OP}+\overrightarrow{PF}=\begin{pmatrix}2\\-3\\0\end{pmatrix}+\begin{pmatrix}1\\2\\4\end{pmatrix}=\begin{pmatrix}3\\-1\\4\end{pmatrix}$ $\therefore F(3,-1,4)$	
(ii)	$\overrightarrow{AB}=\begin{pmatrix}5\\0\\7\end{pmatrix}-\begin{pmatrix}-1\\-12\\4\end{pmatrix}=\begin{pmatrix}6\\12\\3\end{pmatrix}=3\begin{pmatrix}2\\4\\1\end{pmatrix}$ $l_2: \underset{\sim}{r}=\begin{pmatrix}5\\0\\7\end{pmatrix}+\mu\begin{pmatrix}2\\4\\1\end{pmatrix}\quad \mu\in\mathbb{R}$ $\theta=\cos^{-1}\dfrac{\left\|\begin{pmatrix}2\\4\\1\end{pmatrix}\bullet\begin{pmatrix}2\\-3\\0\end{pmatrix}\right\|}{\sqrt{21}\sqrt{13}}=\cos^{-1}\dfrac{8}{\sqrt{21}\sqrt{13}}=61.0^{\circ}\ \text{(1 d.p)}$	
(iii)		

	$\overrightarrow{OD}=\dfrac{\overrightarrow{OA}+2\overrightarrow{OB}}{3}$ or $\overrightarrow{OB}=\dfrac{\overrightarrow{OA}+\overrightarrow{OD}}{2}$ $\overrightarrow{OD}=\dfrac{\begin{pmatrix}-1\\-12\\4\end{pmatrix}+2\begin{pmatrix}5\\0\\7\end{pmatrix}}{3}$ or $\overrightarrow{OD}=2\overrightarrow{OB}-\overrightarrow{OA}$ $\overrightarrow{OD}=\begin{pmatrix}3\\-4\\6\end{pmatrix}$ or $\overrightarrow{OD}=2\begin{pmatrix}5\\0\\7\end{pmatrix}-\begin{pmatrix}-1\\-12\\4\end{pmatrix}=\begin{pmatrix}11\\12\\10\end{pmatrix}$ $\therefore D(3,-4,6)$ or $D(11,12,10)$	
	Since D lies on l_2 and $l_2: \underset{\sim}{r}=\begin{pmatrix}5\\0\\7\end{pmatrix}+\mu\begin{pmatrix}2\\4\\1\end{pmatrix}$ $\mu\in\mathbb{R}$, $\overrightarrow{OD}=\begin{pmatrix}5+2\mu\\4\mu\\7+\mu\end{pmatrix}$ for some μ $\overrightarrow{AD}=(3+\mu)\begin{pmatrix}2\\4\\1\end{pmatrix}$ and $\overrightarrow{BD}=\mu\begin{pmatrix}2\\4\\1\end{pmatrix}$ $\overrightarrow{AD}=2\overrightarrow{DB}$ $\Rightarrow 3+\mu=-2\mu$ $\Rightarrow \mu=-1$ $\overrightarrow{AD}=2\overrightarrow{BD}$ $\Rightarrow 3+\mu=2\mu$ $\Rightarrow \mu=3$ $\therefore D(3,-4,6)$ or $D(11,12,10)$	
	Since D lies on l_2 and $l_2: \underset{\sim}{r}=\begin{pmatrix}5\\0\\7\end{pmatrix}+\mu\begin{pmatrix}2\\4\\1\end{pmatrix}$ $\mu\in\mathbb{R}$, $\overrightarrow{OD}=\begin{pmatrix}5+2\mu\\4\mu\\7+\mu\end{pmatrix}$ for some μ	

	$\overrightarrow{AD}=\overrightarrow{OD}-\overrightarrow{OA}=\begin{pmatrix}6+2\mu\\12+4\mu\\3+\mu\end{pmatrix}$ $\overrightarrow{BD}=\overrightarrow{OD}-\overrightarrow{OB}=\begin{pmatrix}2\mu\\4\mu\\\mu\end{pmatrix}$ $\left\lvert\overrightarrow{AD}\right\rvert=2\left\lvert\overrightarrow{BD}\right\rvert$ $\left\lvert\overrightarrow{AD}\right\rvert^2=4\left\lvert\overrightarrow{BD}\right\rvert^2$ $(6+2\mu)^2+(12+4\mu)^2+(3+\mu)^2=4\left(4\mu^2+16\mu^2+\mu^2\right)$ From GC, $\mu=-1$ or $\mu=3$ $\therefore D(3,-4,6)$ or $D(11,12,10)$	
(iv)	Direction vector of l_3 $=\begin{pmatrix}2\\-3\\0\end{pmatrix}\times\begin{pmatrix}2\\4\\1\end{pmatrix}$ $=\begin{pmatrix}-3\\-2\\14\end{pmatrix}$ $l_3:\underset{\sim}{r}=\begin{pmatrix}-1\\-12\\4\end{pmatrix}+\alpha\begin{pmatrix}-3\\-2\\14\end{pmatrix}\quad \alpha\in\mathbb{R}$	
14 (i)	Interest rate for 12 months = $p\%$ Interest rate per month = $\frac{p}{100}\times\frac{1}{12}$ $\therefore$ Interest before 1 Sept $=\frac{pL}{1200}$	

14 (ii)

Month	Outstanding Loan
0	L
1	$\left(1+\frac{p}{1200}\right)L-x$
2	$\left(1+\frac{p}{1200}\right)\left[\left(1+\frac{p}{1200}\right)L-x\right]-x$ $=\left(1+\frac{p}{1200}\right)^2 L-\left(1+\frac{p}{1200}\right)x-x$
3	$\left(1+\frac{p}{1200}\right)\left[\left(1+\frac{p}{1200}\right)^2 L-\left(1+\frac{p}{1200}\right)x-x\right]-x$ $=\left(1+\frac{p}{1200}\right)^3 L-\left(1+\frac{p}{1200}\right)^2 x-\left(1+\frac{p}{1200}\right)x-x$
n	$\left(1+\frac{p}{1200}\right)^n L-\left(1+\frac{p}{1200}\right)^{n-1} x$ $\ldots-\left(1+\frac{p}{1200}\right)^2 x-\left(1+\frac{p}{1200}\right)x-x$

$L_n=\left(1+\frac{p}{1200}\right)^n L-\left(1+\frac{p}{1200}\right)^{n-1} x$ -------------------------- (*)

$\ldots-\left(1+\frac{p}{1200}\right)^2 x-\left(1+\frac{p}{1200}\right)x-x$

$$=\left(1+\frac{p}{1200}\right)^n L-x\left[\left(1+\frac{p}{1200}\right)^{n-1}+\left(1+\frac{p}{1200}\right)^{n-2}\ldots+1\right]$$

$$=\left(1+\frac{p}{1200}\right)^n L-x\left[\frac{\left(1+\frac{p}{1200}\right)^n-1}{\left(1+\frac{p}{1200}\right)-1}\right]$$

$$=\left(1+\frac{p}{1200}\right)^n L-\frac{1200x}{p}\left[\left(1+\frac{p}{1200}\right)^n-1\right] \text{ (Shown)}$$

14 (iii)

In 30 years, n = 360,

$$504000\left(1+\frac{2.6}{1200}\right)^{360}-\frac{1200x}{2.6}\left(\left(1+\frac{2.6}{1200}\right)^{360}-1\right)\le 0$$

$$\frac{1200x}{2.6}\left(\left(1+\frac{2.6}{1200}\right)^{360}-1\right)\ge 504000\left(1+\frac{2.6}{1200}\right)^{360}$$

$$x \geq \frac{1098534.707}{544.4457028}$$
$$\geq 2017.712145$$

OR

Outstanding loan, Y_1

$$= 504000\left(1+\frac{2.6}{1200}\right)^{360} - \frac{1200x}{2.6}\left(\left(1+\frac{2.6}{1200}\right)^{360} - 1\right)$$

By GC, when $Y_1 = 0$, $x = 2017.712146$

Check: When $x = 2017.71$, $Y_1 = 1.1684474 > 0$

$\therefore$ Their monthly instalment is \$2017.72. (nearest cent)
Or
Their monthly instalment is \$2020. (3 s.f.)

(iv)

When

$$504000\left(1+\frac{2.6}{1200}\right)^{n} - \frac{1200 \times 4000}{2.6}\left[\left(1+\frac{2.6}{1200}\right)^{n} - 1\right] \leq 0,$$

by GC table,

X	Y2
140	28983
141	25046
142	21100
143	17146
144	13183
145	9211.4
146	5231.3
147	1242.7
148	-2755
149	-6761
150	-10775

$\therefore n = 148 \Rightarrow$ 12 years 4 months

The couple will fully repay the loan on 1 December 2033.

Final repayment = \$1245.38